

Draw the radii SA , SB , SC , SD , SE , SF from the centre to every vertex of the path. As the body advances one step, its radius sweeps across a thin triangle: the area cut out between successive radii is precisely the area described "by a radius drawn to the centre." Kepler saw this pattern in the wanderings of Mars — the planet moves fast near the Sun and slow when far, yet always paints equal areas in equal times. Newton's aim is to *explain* it: he will prove that triangle SAB , then SBC , then SCD , and so on are all exactly equal, so that the running total of swept area climbs in perfect lockstep with the clock. In modern language this equal-area law is nothing but the conservation of angular momentum, hiding in plain geometric sight.